and the end of negative half cycle is 360°. The wave consists of four quadrants like a circle.

1.2.2.8 Frequency

The number of complete cycles per second is called Frequency. Its unit is Hertz (Hz). Higher value units are kilo-Hertz (kHz), Mega-hertz (MHz), Gigahertz(GHz) and so on.

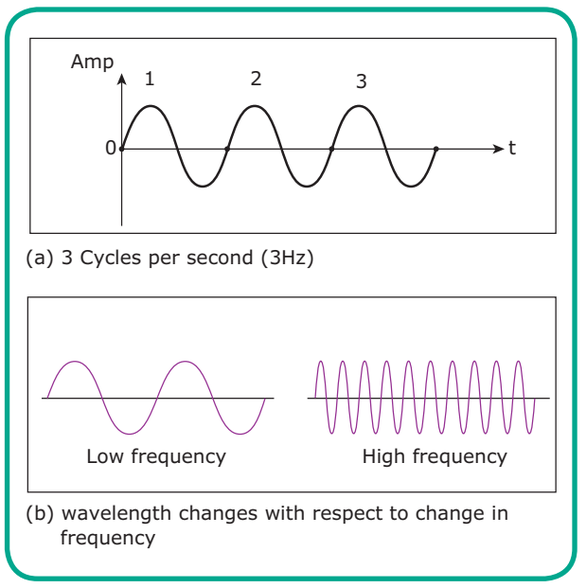


Figure 1.2 Frequency of the AC waveform

If there are 3 full cycles occur in one second, then its frequency is 3Hertz.It is shown in Fig 1.2 (a)

Low frequency has long wavelength and high frequency has short wavelength. It is shown in Fig 1.2 (b)

$$\text{Wavelength } (\lambda) = \frac{\text{Velocity of light or Electromagnetic waves } (C)}{\text{Frequency } (F)}$$

$$\lambda = C / F$$

(Unit of wavelength in meter, velocity of light - constant - 3×10^8 meter/sec and unit of frequency in Hertz)

The unit of frequency was named after German scientist Heinrich Hertz.

1.2.3 Direct Current

The direction of current does not vary with time. Therefore, it is called as Direct Current. We can get DC from battery or by converting AC into DC using rectifier circuits. Figure 1.3 represents the direct current.

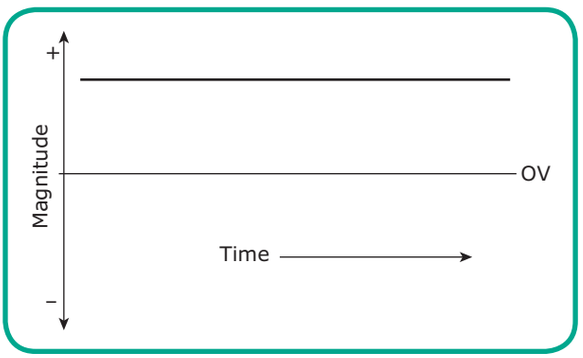


Figure 1.3 Representation of direct current

ACTIVITY 1.1

- Take a cycle dynamo.
- Connect the output of it with a multimeter in DC voltage range.
- Rotate the dynamo.
- Observe the voltage reading in the meter.
- Connect an LED in series with 1 kΩ with the output of the dynamo.

You will see the bulb glowing.

DC has two terminals namely +ve and –ve and hence battery symbol is used to denote DC.

1.3 ELECTRICAL PROPERTIES

Basic Electrical properties are Voltage, Current and Resistance.

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